

myShop: Technical & Operational Documentation

AI-Powered Offline Point of Sale & Inventory Management

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1 Executive Summary

The **myShop** application (formerly Mbeck enterprise) is a sophisticated, offline-first Point of Sale (POS) and inventory management solution. Specifically engineered for retail shops and mobile money (M-Pesa) agents, it addresses the challenges of low-connectivity environments by providing high-speed local data persistence, advanced financial reconciliation, and a pioneering AI-driven visual product recognition engine.

The core value proposition lies in its ability to turn entry-level Android devices, such as those utilizing the Unisoc T615 chipset, into powerful business analytical tools without the need for cloud subscriptions or active internet connectivity.

2 AI Visual Recognition Architecture

The defining feature of myShop is its visual matching engine. Unlike standard barcode scanners that require high-contrast labels, myShop utilizes computer vision to identify products based on their physical branding and packaging essence.

2.1 EfficientNet-Lite0 Feature Extraction

The application utilizes a quantized **EfficientNet-Lite0** model via the TensorFlow Lite (TFLite) framework.

- **Input Processing:** Images are resized to a 224×224 RGB matrix.
- **Vectorization:** The model extracts a 1280-dimension mathematical "feature vector" (embedding) from the penultimate layer.
- **Similarity Metric:** Comparison is performed using **Cosine Similarity**, measuring the angular distance between a live scan and the stored inventory vectors.

2.2 The "Global AI Training" Mechanism

To improve accuracy, the system implements a proprietary *Multi-Perspective Training* strategy. When a user triggers the "Train AI" function in the inventory:

1. The system performs a triple-crop analysis (Center, Top-Left, Bottom-Right) on existing product photos.
2. These perspectives are averaged into a stable "Master Centroid."
3. This process allows the AI to "relearn" branding details on packaging edges that might be missed by a single-center focus.

3 Financial Management & Accounting

myShop provides a dual-layer financial tracking system designed to eliminate "lost money" through rigorous reconciliation and profit analysis.

3.1 The Profit Engine

The profit logic is calculated at the transaction level using the formula:

$$\text{Net Profit} = (\text{Selling Price} - \text{Applied Discount}) - \text{Original Buying Cost}$$

By joining the `txn` (transaction) table with the `inventory` table, the application generates real-time reports on the most profitable items, rather than just those with the highest volume.

3.2 Accounting Reconciliation & Disparity Tracking

The Accounting Screen serves as a digital ledger for shop owners to balance physical cash and digital balances across multiple providers (M-Pesa, Co-op, Equity, Airtel).

- **Mathematical Input:** Fields support inline addition (e.g., 5000 + 1200) to help users sum up totals without leaving the app.
- **Disparity Analysis:** The system compares the "Expected Total" (Last Count + Sales since last count) against the "Actual Count" provided by the user, immediately highlighting surpluses or shortages.

4 Security & Anti-Fraud Systems

Despite being an offline application, myShop implements enterprise-grade security for customer interaction.

4.1 Digital Receipt Verification (Version 10)

To prevent customers from presenting fake "Payment Successful" WhatsApp messages, myShop generates a unique **Security Signature** for every sale.

- **Digital Fingerprint:** During a sale, a signature is generated using the transaction timestamp and total amount, which is then permanently logged in the database.
- **The Verification Tool:** Employees can use the "Verify Receipt" tool on the home screen to input a receipt's Ref Code and Signature. The system performs a direct SQL lookup to confirm if the details match the secure logs.

5 Technical Infrastructure

The application is built on the Flutter framework, prioritizing low-latency performance and cross-version compatibility.

5.1 Database Schema (SQLite)

The database utilizes `sqflite` with a versioned migration path. Current schema highlights include:

- **inventory:** Stores primary product data and the master visual vector.

- **inventory_vectors:** A memory table that stores "Secondary Perspectives" for items, allowing the AI to remember how an item looks under different lighting.
- **txn:** A high-indexed table for rapid history retrieval.

5.2 Performance on Low-End Hardware

Special consideration was given to the Unisoc T615 chipset:

- **EMA Momentum:** Exponential Moving Averages (EMA) with a momentum of 0.95 are used for online vector refinement.
- **Batch Restraint:** During AI training, the system implements a 50ms artificial delay between items to prevent thermal throttling.

6 User Operational Workflows

6.1 Adding New Inventory

1. Capture a clear, top-down photo of the product.
2. Enter the name, buying price, and selling price.
3. The AI will automatically generate the initial visual signature.
4. (Recommended) Use the "Train AI with new Angle" button to capture the back or side of the package for better recognition.

6.2 Performing a Quick Sell

1. Open the scanner.
2. Point the camera at the product.
3. A vertical list of the top 5 AI matches will appear.
4. Select the item, apply a discount if necessary, and "Proceed."

6.3 Daily Closing

1. Use the "Cash Count" tool to count physical notes and coins.
2. Save the cash total to the Accounting module.
3. Input digital balances (M-Pesa, Bank).
4. Review the Disparity Report to ensure the shop is "Balanced."

7 Conclusion

myShop represents a bridge between traditional retail and modern AI technology. By keeping all data local and using advanced mathematical recognition, it provides shop owners with a level of control and security previously reserved for high-end, cloud-integrated ERP systems.